Yashashwi Singhania

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Education

Indian Institute of Technology (BHU) Varanasi — Aug 2023 – May 2028 (Expected)
B.Tech. + M.Tech. in Biochemical Engineering | CGPA: 8.74 — *Varanasi, India*

Experience

Machine Learning Intern — June 2025 – July 2025
Fxis.ai — *Remote*

- Architected a **multimodal RAG pipeline** utilizing **Weaviate** to parse complex enterprise documents.
- Engineered **advanced embedding models** to seamlessly extract **structured text and tabular data**.
- Automated **analytical report generation** from UK corporate tax filings to **accelerate compliance**.

Projects

Arras - *Swift, SwiftUI, AppKit, Core Animation* — *June 2026*

- Developed a native, open-source **macOS desktop widget** using **SwiftUI and AppKit** that preserves true image aspect ratios without forced cropping, shipped to **1,000+ downloads**.
- Engineered a custom "Smart Canvas" engine utilizing **GPU-accelerated Core Animation** crossfades and cyclic spatial memory, ensuring the retention of independent coordinate and proportional size states for dynamically rotating user images.
- Implemented deep macOS system integrations: **non-activating NSPanels** so widgets never steal focus, a four-level window stack derived from **CGWindowLevelForKey**, and cross-application edge snapping through **CGWindowListCopyWindowInfo** that needs no accessibility permissions. | **Links:** GitHub | Download

OMR-Checker - *Python, OpenCV, NumPy* — *May 2026*

- Built an **optical mark recognition (OMR)** system using **Python and OpenCV** to process **200+ sheets per minute**.
- Achieved **near 100% accuracy** on scanned documents and **90% on mobile images** under varying lighting conditions.
- Engineered **dynamic layout configurations** and **visual debugging tools** to adapt to complex custom OMR templates. | **Links:** GitHub

TDigest - *C++, Multithreading, STL* — *June 2025*

- Built C++17 **t-Digest library** achieving $\mathcal{O}(\delta)$ space complexity for **real-time streaming quantile estimation**.
- Optimized **multithreaded ingestion system** achieving $8.4M+$ **points/second throughput** with $0.52\mu s$ **query latency**.
- Created **automated benchmarking framework** with build scripts and **performance profiling** across multiple configurations. | **Links:** GitHub

pyFOOMB GUI - *Next.js, React, Data Visualization* — *May 2026*

- Built a responsive **Next.js GUI** for pyFOOMB to streamline complex **bioprocess modeling workflows**.
- Engineered **interactive visualizations** and dynamic layouts for **real-time multi-reactor analysis**. | **Link:** GitHub

Campus Involvement

Club of Programmers (COPS), IIT BHU
Category Head

- Organised Byte the Bits under Technex, attracting **over 3000 participants** all over India.
- Facilitated an 'Intro to Python' workshop attended by **200+ students** at IIT BHU.

Achievements

- **Programming & Hackathons:** Achieved a **Codeforces rating of 1720** (`hackerman15`), awarded **1st Prize in DevBits** at Udyam (IIT BHU 2025), and secured **top 50 national rankings** in AMS Derive 2026 (**AIR 13** in POSTERIOR, **AIR 39** in PRIOR).
- **National Examinations & Olympiads:** Attained **AIR 1105 in KVPY-SA** 2021-2022, secured **State Rank 20 (Uttar Pradesh)** in the **National Talent Search Exam (NTSE)**, and qualified the **Indian Olympiad Qualifier in Mathematics (IOQM)** 2021.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, Markdown, LaTeX
Web/Frameworks: NextJS, React, Flask, Tailwind CSS, SwiftUI, AppKit
Tools/Platforms: Docker, Git, GitHub Actions, Render, Google Colab, VS Code, Vercel, Bash, Zsh, SSH
Operating System: Linux, macOS